APPLICATION PORTFOLIO

Niklas

Bubeck

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LANGUAGES

German:
Native proficiency

English:
Full professional proficiency

French:
Limited working proficiency

Spanish:
Limited working proficiency

SKILLS - Technical

MS Office
Git
Docker
Jira
Confluence

C++
C
Python
SQL

PyTorch
TensorFlow
ROS
Linux

PROFILE

Detail-oriented master's student with multiple years of experience in the field of autonomous driving, automated intelligent systems, computer vision, artificial intelligence as well as IoT. Showed outstanding leadership skill while managing the driverless project and lead the autonomous software team of an accelerated and successful FSAE team. Interdisciplinary and dedicated to tech, having worked in vastly different environments from aggressive start-ups to Germanys biggest corporations.

EDUCATION

Technische Universität München | October 2020 - October 2023
Master of Science: Robotics Cognition Intelligence
Thesis: An attention-based image denoising network leveraging information of both spatial and frequency domain

Friedrich Alexander Universität Erlangen | October 2016 - October 2020
Bachelor of Science: Medical Engineering (Pattern Recognition/Imaging)
Thesis: Truncation-correction Method for X-ray Dark-field Computed Tomography

WORK EXPERIENCE

TUM.ai
Head of Industry Department | February 2023 - Now
The industry department is focusing on connecting motivated students with experience in ML/AI fields with companies who are seeking to introduce ML/AI solutions to their ongoing products.

- Leading the Industry Department with ~20 members
- Leading and designing the TUM.ai industry phase with an estimated 130k€ annual turnover
- Product Management and Strategy

Industry Department Active Member | November 2022 - February 2023

- Participation in AI projects & workshops
- Partner communication & project sourcing
- Supervising teams as AI expert

IAV AG
Working Student | November 2021 - December 2022
Working student position in the field of Hardware in the Loop Test automatization of Advanced Driver-Assistance Systems for BMW and Cariad.

- Implementation of windows services, watchdogs, and servers for automatic uploads of the test benches - *Python, C, Databases*
- Developing web extensions (Chrome, Firefox), which allow easier handling of the testing processes - *JavaScript, HTML5, CSS, HTTP, REST*
- Runtime optimization using multithreading and multiprocessing solutions - *Python, C, Networking*
- Maintenance of the test bench and relevant software modules

Siemens AG
Working Student | April 2019 - September 2021
Working student position covering the setup and execution of software as well as hardware tests for varies projects, including energy storage systems, cloud platforms and multiple IoT devices.

- Developing UI and E2E tests for the Android and iOS application - *Kotlin, Java, Espresso, Swift*
- Programming and testing of embedded systems and IoT devices - *Python, C++, MQTT*
- Multiple small-scale developments for different areas such as web and mobile applications, including initial experiences with cloud systems.

SKILLS - Professional

- Problem Solving
- Team Building
- Meeting Deadlines
- Punctuality
- Prioritization
- Organization
- Leadership

AWARDS

2nd Place (out of 90) in the Pattern Recognition Lab Deep Learning Challenge "Classification of Solar Cell Defects"

RELEVANT COURSES

- Computer Vision: Multiple View Geometry
- Introduction to Deep Learning
- Introduction to Machine Learning
- Visual Data Analytics
- Bio inspired data processing and learning with neural networks
- Pattern Recognition

WORK EXPERIENCE

Formula SAE - High-Octane Motorsports e.V.:
During this time, I was able to gain a lot of experience in different areas. I joined the team as a software developer when we were starting to develop our first autonomous racecar. Therefore, I had the honor to make many experiences of different development stages. Beginning from a modified go kart up to a fully autonomous racecar.
Advisory Software Engineer | Oktober 2020 - September 2021
Advisor and developer in the final year of the FSAE experience.

- Interaction with student initiatives and university affairs
- Advisor for: clean code, evaluating technological concepts of software as also hardware stacks
- Implementations regarding model predictive control

Project lead Driverless, Team lead autonomous Software | August 2019 - September 2020
In addition to software development, taking on managing tasks and leadership role:

- Leading the Driverless team with ~30 and Software team with ~10 members.
- Buildup of test-driven development and continuous integration as well as agile process management.
- Design and development of: linear-quadratic regulators with dynamic non-linear vehicle models - *C++, Python*
- Pathfinding and motion planning, trajectory optimization. Using triangulation, probabilistic and filtering methods - *C++, Python*
- Co-developing in the fields of SLAM and Perception. Using FastSLAM methods and YOLO as well as other single shot multibox detectors. - *C++, Python*
- System integration of sensor technology - Radar, LiDAR, IMU, Cameras

Software Developer | October 2018 - August 2019

- Developing motion planning methods mainly using splines and clothoids - *C++*
- Assistance in manufacturing of the wiring harness including the communication system of the vehicle - *CAN bus, Ethernet*
- Assistance in the production of turned parts as well as mechanical components and assemblies

Chair of Technical Electronics (LTE) FAU
Student Assistant | November 2018 - February 2019
Student assistant holding the tutorial for 'fundamentals of electrical engineering'. Teaching other students and preparing them for the exam.

- First contact with teaching in a university setting
- More detailed study about electrical engineering

VOLUNTEERING

Mentoring
Rock Your Life! München e.V. | December 2022
Rock Your Life is a social organization that pairs university students with school pupils to provide mentorship and guidance. The university students act as mentors to the school pupils, offering advice and support on a variety of topics, including life and career choices. The goal of the organization is to empower and inspire young people to reach their full potential and make a positive impact in their communities.

Web development: dein-aufgabenblatt.de
Project | Januar 2018 - Dezember 2018
Design and development of a learning platform for students using Ruby on Rails. Some key features of the web application are: - *Ruby on Rails, Ruby, JavaScript, MySQL*

- User Authentication, Adminportal
- Rating and ban systems
- Notification Mailer and In-app notifications
- Comment and Forum System
- LaTex integration